Mathematical floor

Unit 0 · reference sheet · AI/ML in 35 topics

Multiplication: the only operation

$x \cdot w = \sum_i x_i w_i = \|x\|\|w\| \cos\theta \quad \cos\theta = \dfrac{x \cdot w}{\|x\|\|w\|} \in [-1, 1]$ $(m \times n)(n \times k) \to (m \times k)$ inner dims must match

$Wx = m$ dot products at once; $xW \neq Wx$

Derivatives

$\dfrac{d}{dw} w^n = nw^{n-1} \qquad \dfrac{d}{dw}(aw + b) = a$

Chain: $\dfrac{d}{dw} f(g(w)) = f'(g(w))g'(w)$

Update: $w \leftarrow w - \eta \nabla f$ (minus: gradient is uphill)

Probability

$E[X] = \sum_x x\, p(x) \qquad \text{Var} = E[(X - EX)^2]$

$L(\theta) = p(\text{data} \mid \theta) \qquad \text{NLL} = -\sum_i \log p_\theta(y_i \mid x_i)$

$\text{softmax}(z)_i = \dfrac{e^{z_i}}{\sum_j e^{z_j}} \qquad \text{perplexity} = e^{\text{loss}}$

Gotchas

Symptom	Correction
Big dot ⇒ similar	Length leaks in. Use cosine when norms are noise.
Gradient points downhill	It points uphill; subtract it.
η too large	Oscillates: $w = 3, \eta = 1$ on w^2 gives -3 forever.
Likelihood = $P(\theta \mid D)$	It is $P(D \mid \theta)$.

This unit at a glance

#	Topic	You can
1	Vectors, matrices, dot product	read any score function as a dot product and check shapes mechanically
2	Slope → gradient	compute a gradient and explain why training steps against it
3	Probability, expectation, likelihood	explain why a loss is a negative log-likelihood; read mean/variance off a dist

If you see this → do that

Signal	Move
Compare two vectors, norms vary	cosine
Need a probability	sigmoid / softmax
Predict a number	squared error
Predict a class	BCE / cross-entropy
Shapes disagree	write inner dims

Numbers worth memorising

$\cos \in [-1, 1]$; perpendicular ⇒ 0; 45° ⇒ 0.707

$0.25^{10} = 9.5 \times 10^{-7} \qquad 1.5^{10} = 58$

$e^3 \approx 20.1 \qquad \ln 2 \approx 0.693$

$a^\top a = \|a\|^2 \qquad (AB)^\top = B^\top A^\top$